

# State of the Field: Year 1 of the PWM Low-Dose CT Challenge

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(will be re-cut annually as new yearly editions; this draft is the Year-1 template.)

## Abstract

**Background.** Low-dose CT reconstruction has accumulated years of method papers without producing a citation standard against which longitudinal progress can be measured. A practitioner deciding which method to deploy on a new vendor’s data, or a radiologist asking which dose reduction is now diagnostically safe for a specific clinical task, has no shared reference.

**Methods.** The PWM Low-Dose CT Challenge is a permanent, content-addressed, citation-standard competition hosted on the PWM blockchain registry, scored against the multi-vendor PWM-LDCT benchmark (to be confirmed, 2027a) using the signal-equivalence framework of (to be confirmed, 2027b). Submissions take the form of sandboxed Docker RunBundles whose published results are re-executed and bit-identically verified by the scoring service. We review the first year of the challenge: [TODO: N submissions] community entries from [TODO: N institutions] institutions in [TODO: N countries] countries, including [TODO: N] industrial vendor entries.

**Findings.** We characterize Year-1 submissions across five axes: reconstruction quality (PSNR / SSIM / LPIPS), downstream task performance (lung-nodule detection AUC), cross-vendor generalization, signal-equivalence credential coverage at multiple dose ratios, and open-source license posture. Three findings: (i) the field has converged on [TODO: trend]; (ii) credential coverage at  $r = 0.25$  reached [TODO: X%] but  $r = 0.10$  remains an open problem; (iii) cross-vendor generalization remains the largest unsolved gap, median PSNR drop [TODO: X dB].

**Significance for clinical practice.** For the first time, a radiologist or physicist can consult Table 3 to look up the most aggressive open-source-credentialed dose reduction for a specific task and subpopulation. This paper inaugurates an annual review series (to be confirmed, 2028) that will recur each year, citing every submission from that year and providing a longitudinal snapshot of low-dose CT reconstruction as the field evolves.

## 1 Introduction

The deep-learning era of low-dose CT reconstruction has produced a steady stream of incremental method papers without producing a citation standard. A new method published in 2023 typically reports a PSNR improvement of fractions of a decibel over the previous best, on the same dataset (the Mayo Clinic AAPM 2016 cohort (AAPM, 2016)), using the same evaluation metrics, with little or no cross-vendor evaluation, no per-pixel uncertainty quantification, and a license that varies

from Apache 2.0 to “available on request.” A practitioner picking a method to deploy on data from a different scanner has no clear choice; a regulator evaluating an FDA submission has no shared reference; a researcher writing a follow-on paper cannot tell whether the previous-best number is from a comparable evaluation protocol.

The PWM Low-Dose CT Challenge addresses these structural gaps by combining three artifacts that are individually well-established but rarely co-located: a **multi-vendor, paired-dose, on-chain-anchored benchmark dataset** (to be confirmed, 2027a); a **probabilistic, task-conditional, modality-general signal-equivalence framework** (to be confirmed, 2027b); and a **permanent competition** whose submissions, evaluation, and scoring are reproducible end-to-end via PWM RunBundles (PWM Protocol, 2026). Submissions take the form of a sandboxed Docker container with a pinned eval entry point and a published `results.json`; the challenge’s scoring service re-executes the container, verifies the published results against the live execution, and posts the verified score to a public leaderboard. The L3 dataset spec and the L2 framework spec are content-addressed on the PWM blockchain, so a result tied to `pwm_l3_spec:pwm_low_dose_ct_benchmark:1.0.0` and `pwm_l2_spec:dose_equivalence_v1:0.1.0` is bit-identically reproducible across years even as the dataset is later extended.

This paper is the inaugural annual review of the challenge. It is the first of an intended sequence: each year, an analogous paper will summarize the cohort of new submissions, characterize cross-method trends, perform clinical-impact analysis (which dose levels are now diagnostically equivalent for which subpopulations), and identify the open problems that should define the next year’s work. The recurring annual structure — with consistent benchmark, framework, and reporting standards across years — is the mechanism by which the field can finally measure longitudinal progress.

**What makes this review series distinctive.** Reviews of low-dose CT reconstruction have appeared periodically (Wang et al., 2020), but they share three limitations that the present series is structurally designed to avoid:

1. **Static snapshots.** Prior reviews are written once. A method published two years later is not visible until the next ad-hoc review. The PWM annual cadence guarantees a yearly cohort summary.
2. **Anchorless evaluation.** Prior reviews compare numbers reported on differently-cleaned versions of the same dataset, with no mechanism to verify that those numbers were produced under comparable evaluation protocols. The PWM challenge ties every cited number to a content-addressed benchmark hash and a content-addressed framework hash, so cross-year comparisons are bit-identically reproducible.
3. **Pixel-wise metrics only.** Prior reviews emphasize PSNR / SSIM. The PWM challenge requires every submission to report a 5-tuple credential under the signal-equivalence framework (to be confirmed, 2027b), which ties the comparison to a specified clinical task, subpopulation, margin, and confidence level.

### Contributions of this Year-1 review.

1. A summary of the year-1 submission cohort across methodology classes, institutions, and reported metrics (Section 3).
2. Cross-method trends across reconstruction quality, downstream task performance, cross-vendor generalization, and uncertainty quantification (Section 4).

3. Clinical-impact analysis: which dose levels are now signal-equivalent under the framework of (to be confirmed, 2027b) for which clinical tasks (Section 5).
4. A named list of open problems we believe define Year 2 (Section 6).
5. A protocol for the year-2 challenge cycle (Section 7).

## 2 The Challenge

### 2.1 Benchmark dataset (L3 spec)

The challenge uses the PWM-LDCT dataset (to be confirmed, 2027a): **[TODO: N patient scans]** paired-dose CT acquisitions across **[TODO: N vendors]** vendor platforms at **[TODO: N sites]** clinical sites. Each scan provides DICOM raw projections, image-space reconstructions at four dose levels (10%, 25%, 50%, 100%), and majority-vote radiologist annotations for lung-nodule detection and liver-lesion segmentation tasks. The dataset version evaluated in Year 1 is content-addressed at `pwm_l3_spec:pwm_low_dose_ct_benchmark:1.0.0`.

### 2.2 Evaluation framework (L2 spec)

Performance is reported through the signal-equivalence framework of (to be confirmed, 2027b): every submission must declare its 5-tuple credentials of the form  $(r, T, \varepsilon, \alpha, \Pi)$  for at least one canonical clinical task. The framework version evaluated in Year 1 is content-addressed at `pwm_l2_spec:dose_equivalence_v1:0.1.0`. Pixel-wise metrics (PSNR / SSIM / LPIPS) are reported as standard quality measures but are not the basis for leaderboard ranking; the leaderboard ranks by task-AUC on the canonical chest lung-nodule detection task.

### 2.3 Submission protocol

A submission is a PWM RunBundle: a content-addressed bundle containing a pinned Docker image, an eval entry point, a published `results.json`, and a corresponding 5-tuple credential JSON. The scoring service executes the RunBundle in a sandboxed container on the PWM-LDCT test split, verifies that the published `results.json` matches the live execution (within IEEE 754 FP tolerance), and posts the verified score. Failure to verify triggers an automatic rejection with a public failure-reason record. Tampered submissions are slashed via the standard PWM credential-violation protocol.

### 2.4 Prize structure

The Year-1 prize pool is **[TODO: N PWM tokens]** (\$**[TODO: N USD equivalent]**), distributed top-10:

| Place    | Share                     |
|----------|---------------------------|
| 1st      | 25%                       |
| 2nd      | 15%                       |
| 3rd      | 10%                       |
| 4th–10th | 50% / 7 $\approx$ 7% each |

We chose top-10 distribution over winner-take-all to encourage broad participation. Participation rates are reported in Section 3.

### 3 Year-1 cohort

**Submission counts.** We received  $N_{\text{sub}} = [\text{TODO: } ]$  community submissions between the challenge launch and the Year-1 close date. Of these,  $[\text{TODO: } N]$  were from academic institutions,  $[\text{TODO: } N]$  from industrial vendors, and  $[\text{TODO: } N]$  from independent or unaffiliated submitters.

**Methodology classes.** Submissions classified by architecture family in Table 1  $[\text{TODO: populate}]$ . Trend:  $[\text{TODO: e.g. diffusion-based reconstructions overtook transformer-based reconstructions in count}]$ .

| Architecture family              | Count              |
|----------------------------------|--------------------|
| Unrolled iterative               | $[\text{TODO: } ]$ |
| Pure CNN (incl. encoder-decoder) | $[\text{TODO: } ]$ |
| Transformer-based                | $[\text{TODO: } ]$ |
| Diffusion / score-based          | $[\text{TODO: } ]$ |
| Hybrid / other                   | $[\text{TODO: } ]$ |
| <b>Total</b>                     | $[\text{TODO: } ]$ |

Table 1: Year-1 submission distribution by architecture family.

**Verification statistics.** Of the  $[\text{TODO: } ]$  submissions,  $[\text{TODO: } ]$  passed bit-identical re-execution verification on first attempt;  $[\text{TODO: } ]$  required resubmission to fix reproducibility issues;  $[\text{TODO: } ]$  were rejected for unrecoverable verification failure. The most common verification failure was  $[\text{TODO: e.g. missing CUDA RNG seed propagation}]$ .

**License posture.**  $[\text{TODO: } N]$  submissions were released under Apache 2.0 or MIT licenses;  $[\text{TODO: } N]$  under non-commercial or research-only licenses;  $[\text{TODO: } N]$  did not declare a license (which is itself a leaderboard policy violation, flagged for the submitter).

## 4 Cross-method trends

### 4.1 Reconstruction quality

PSNR / SSIM / LPIPS distributions are shown in Figure 1. Median PSNR at  $r = 0.25$  across the cohort is  $[\text{TODO: } X \text{ dB}]$ ; the top-3 submissions achieve  $[\text{TODO: } Y \text{ dB}]$ . The leaderboard top entry was  $[\text{TODO: submitter / method name}]$  at  $[\text{TODO: } Z \text{ dB}]$ .

$[\text{TODO: FIGURE: Year-1 cohort PSNR / SSIM / LPIPS distributions on PWM-LDCT test split, per dose level}]$

Figure 1: Year-1 cohort PSNR / SSIM / LPIPS distributions on PWM-LDCT test split, per dose level (placeholder; populate when data lands).

## 4.2 Downstream task performance

Lung-nodule detection AUC at fixed FPR = 0.1 is summarized in Figure 2. We observe that PSNR and task-AUC are [TODO: strongly / weakly] correlated, with a Spearman  $\rho$  = [TODO: X] across the submission cohort. Notably, the top-PSNR submission is [TODO: not / also] the top-task-AUC submission, illustrating that pixel-wise metrics are not interchangeable with clinical-task metrics.

[TODO: FIGURE: Year-1 cohort lung-nodule detection AUC; scatter vs PSNR illustrating the pixel-wise vs task-metric gap]

Figure 2: Year-1 cohort lung-nodule detection AUC; scatter vs PSNR illustrating the pixel-wise vs task-metric gap (placeholder; populate when data lands).

## 4.3 Cross-vendor generalization

A subset of [TODO: N] submissions reported cross-vendor results on the PWM-LDCT leave-one-vendor-out split. The median PSNR drop on the held-out vendor was [TODO: X dB]; the best cross-vendor submission had a drop of only [TODO: Y dB] ([TODO: submitter / method]). The remaining [TODO: N] submissions did not report cross-vendor results, which the leaderboard now records as a missing optional metric.

## 4.4 Uncertainty quantification

[TODO: N submissions] reported per-pixel uncertainty; of these, [TODO: N] provided calibration metrics (Spearman  $\rho$  between predicted variance and observed error). The median calibration  $\rho$  was [TODO: X]. UQ remains underreported despite being one of the most clinically actionable supplementary outputs — a recurring observation in Year 1 that we expect to inform Year-2 submission guidelines.

## 4.5 Signal-equivalence credential coverage

Per the framework of (to be confirmed, 2027b), every submission was required to report at least one 5-tuple credential at  $r = 0.25$ . The distribution of credential verdicts is in Table 2. The  $r = 0.10$  regime — a more aggressive dose reduction — is currently unsolved by most submissions, with only [TODO: N%] of credentials passing at that level.

[TODO: TABLE: Year-1 5-tuple credential verdict distribution (PASS / FAIL / INDETERMINATE) per dose ratio and per canonical task]

Table 2: Year-1 5-tuple credential verdict distribution (PASS / FAIL / INDETERMINATE) per dose ratio and per canonical task (placeholder; populate when data lands).

# 5 Clinical-impact analysis

We summarize, for each canonical clinical task in the challenge, the most aggressive dose-reduction ratio at which at least one Year-1 submission achieves a passing signal-equivalence credential, broken down by patient subpopulation:

| Task                                    | Subpopulation                    | Best passing $r$ | Method   |
|-----------------------------------------|----------------------------------|------------------|----------|
| Lung nodule detection $\geq 5\text{mm}$ | Adult chest, lung screening      | [TODO: ]         | [TODO: ] |
| Lung nodule detection $\geq 5\text{mm}$ | Adult chest, oncology FU         | [TODO: ]         | [TODO: ] |
| Liver lesion segmentation               | Adult abdomen                    | [TODO: ]         | [TODO: ] |
| Liver lesion segmentation               | Adult oncology, post-chemo       | [TODO: ]         | [TODO: ] |
| Body habitus quantification             | Pediatric chest (Year-2 stretch) | N/A              | —        |

Table 3: Most aggressive signal-equivalence-certified dose-reduction ratio by clinical task and subpopulation. “Best passing  $r$ ” is the smallest  $r$  at which at least one Year-1 submission reported a passing ( $r, T, \varepsilon = 0.02, \alpha = 0.05, \Pi$ ) credential.

The clinical translation: a radiologist or physicist reading this table can map to the smallest dose at which their specific task (and their specific subpopulation) is currently supported by an open-source, cross-validated, signal-equivalence-credentialed reconstruction. Year-1 limitations are documented in Section 6.

## 5.1 Implications for clinical practice

**For radiologists.** The verdicts in Table 3 are not regulatory clearance. They are evidence, in the same sense that a published reader study is evidence, that a specific open-source method, applied to the specified subpopulation and task, has been shown under bootstrap inference to be within  $\varepsilon = 0.02$  AUC (or analogous metric) of the named full-dose reference, with 95% confidence. Radiologists considering whether to lobby for protocol changes at their institution can cite specific 5-tuples rather than vendor marketing percentages.

**For medical physicists.** The cross-vendor results in Section 4 provide a starting point for institution-specific QA. A method that achieves a passing credential when trained on the vendor a hospital uses, but a failing or indeterminate credential when trained on a different vendor and tested on this one, is a concrete starting point for the institution’s commissioning plan.

**For institutional review boards.** A protocol that proposes to reduce CT dose for a specified screening cohort can cite a passing 5-tuple credential as preliminary safety evidence for the diagnostic equivalence claim, replacing the present practice of citing vendor white papers with unspecified margins.

**For regulators.** The challenge produces externally-verifiable, content-addressed evidence in a format aligned with the FDA’s emerging guidance on AI/ML-based SaMD subgroup reporting (U.S. Food and Drug Administration, 2024). A reviewer evaluating a submission can request the L4 cert hash and independently re-execute the published evaluation pipeline.

## 6 Open problems for Year 2

We name three open problems we believe will define Year 2 of the challenge:

**Open Problem 1: Pediatric subpopulations.** The Year-1 dataset is adult-only. Pediatric chest CT carries the highest per-scan radiation risk per unit dose and is where dose reduction matters most (Larson et al., 2018). The Year-2 dataset extension will add a pediatric chest subpopulation, and we call on the community to submit methods specifically trained or fine-tuned for it.

**Open Problem 2: Cross-vendor at  $r = 0.10$ .** Methods that achieve passing credentials at  $r = 0.25$  on the trained vendor often fail at  $r = 0.10$  on a held-out vendor. We hypothesize this is a noise-statistics distribution-shift problem rather than an architecture problem; Year-2 submissions are encouraged to explicitly model the noise distribution across vendors.

**Open Problem 3: Real-time inference for ER deployment.** The Year-1 top submissions average [TODO: X seconds per slice] on an A100; emergency-radiology deployment requires [TODO: Y seconds]. This is fundamentally an architecture-choice / quantization / distillation problem and is open for Year 2.

## 7 Year-2 challenge protocol

The Year-2 challenge cycle opens on [TODO: date] and closes on [TODO: date]. Changes from Year 1:

1. **Pediatric subpopulation added.** The dataset is extended to L3 spec v1.1.0; submissions report credentials for both adult and pediatric subpopulations.
2. **Cross-vendor reporting becomes mandatory.** Every submission must report at least one cross-vendor leave-one-out result.
3. **Inference-latency reporting is added to the leaderboard.** Latency is not a tiebreaker for the headline rank, but it is reported alongside performance metrics so users can pick by deployment constraint.
4. **Industrial vendor invitation extended.** [TODO: N vendors] were invited to Year 1; the Year-2 invitation list expands to all [TODO: N vendors] major CT manufacturers.

## 8 Discussion

This paper inaugurates a publication mechanism that did not exist in the low-dose CT field before the launch of the PWM Challenge: a recurring, citable, peer-reviewed annual review whose content is reproducibly tied to a content-addressed benchmark and a content-addressed evaluation framework. Earlier surveys of low-dose CT reconstruction (Wang et al., 2020) have provided invaluable orientation to the field, but they are static snapshots; they cannot tell a reader whether a method published two years after the survey would still be ranked first on the same benchmark using the same evaluation protocol. The recurring annual review closes this gap.

We emphasize three structural properties of the challenge that we believe other medical-imaging fields should consider replicating.

**Content-addressed benchmarks.** A benchmark version that can be silently re-extended, re-split, or re-cleaned between submissions makes year-over-year performance comparisons impossible. Anchoring the dataset version to a cryptographic hash registered on a public blockchain — as the PWM Low-Dose CT Challenge does — eliminates a class of benchmark-drift artifacts that have undermined longitudinal comparison.

**Probabilistic, task-conditional evaluation.** A scalar PSNR or SSIM cannot capture the clinical question — is the reconstruction good enough for the radiologist to make the diagnostic decision? — and the gap between pixel-wise metrics and task performance is well-documented (Kim et al., 2019). The signal-equivalence framework of (to be confirmed, 2027b) provides a reporting standard that ties every claim to a specific task, subpopulation, margin, and confidence.

**Re-executable submissions.** Bit-identical re-execution of every submission, verified by an automated scoring service, eliminates a separate class of reproducibility failures (cherry-picked random seeds, version-skew between training and reported eval, etc.). Adoption of this standard across medical-imaging benchmarks would substantially raise the field’s reproducibility floor.

## Acknowledgments

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## Author contributions

[TODO: Fill at submission per CRediT taxonomy.]

## Competing interests

[TODO: Declare at submission.]

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